

INTERNATIONAL STANDARD

**Touch and interactive displays –
Part 1-2: Generic – Terminology and letter symbols**





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INTERNATIONAL ELECTROTECHNICAL COMMISSION

TOUCH AND INTERACTIVE DISPLAYS –

Part 1-2: Generic – Terminology and letter symbols

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International Standard IEC 62908-1-2 has been prepared by IEC technical committee 110: Electronic display devices.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
110/886/FDIS	110/903/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all the parts in the IEC 62908 series, under the general title *Touch and interactive displays*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

TOUCH AND INTERACTIVE DISPLAYS –

Part 1-2: Generic – Terminology and letter symbols

1 Scope

This part of IEC 62908 provides a list of terminological entries that are frequently used in the literature related to touch and interactive displays in the IEC 62908 series. Terms for various input methods such as touch, hovering, proximity, gesture, eye tracking, and motion recognition are included. This document is applicable to touch displays, interactive displays and their components.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

3.1 Classification of terms

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

Terms for touch and interactive displays are classified as follows (see also Annex A):

- touch displays (3.2);
- components of touch displays (3.3);
- structure types of touch displays (3.4);
- sensor types of touch displays (3.5);
- input operations of touch displays (3.6);
- measurement elements of touch displays (3.7);
- measurement performances of touch displays (3.8);
- interactive displays (3.9);
- components of interactive displays (3.10);
- sensor types of interactive displays (3.11);
- input operations of interactive displays (3.12);
- actuator types of interactive displays (3.13);
- feedback operations of interactive displays (3.14); and
- measurement performances of interactive displays (3.15).

3.2 Touch displays

3.2.1

touch display

touch display module

electronic device that allows the users to input by using touch objects corresponding to the displayed information

3.3 Components of touch displays

3.3.1

touch sensor

sensor which recognizes input through touch

3.3.2

touch display panel

combination of a touch sensor module and a display panel

3.3.3

touch sensor module

touch screen panel

combination of a touch sensor and touch controller that creates an integrated module capable of sensing and processing touch

Note 1 to entry: A cover window is sometimes attached to the touch sensor.

3.3.4

test pattern cell

test sample with patterned electrodes that is made for reliability measurement

3.3.5

cover window

transparent window that covers and protects the touch screen panel and display panel

3.3.6

touch controller

integrated circuit that converts analogue input signals to digital output signals in the touch screen panel

3.3.7

receive electrode array

Rx

array of electrodes used for receiving the electrical signal

3.3.8

transmit electrode array

Tx

array of electrodes with changing voltage used for transmitting the electrical signal

3.4 Structure types of touch displays

3.4.1

add-on type

out-cell type

structure that has touch sensors which are separately fabricated and attached to the display panel

3.4.2**in-cell type**

structure that has touch sensors which are integrated inside the display panel

3.4.3**on-cell type**

structure that has touch sensors which are integrated on the outside of the display panel

3.4.4**hybrid-cell type**

structure that has touch sensors which are both inside and outside the display panel

3.5 Sensor types of touch displays**3.5.1****capacitive touch sensor**

sensor for detecting the change of capacitance

3.5.2**resistive touch sensor**

sensor for detecting the change of resistance

3.5.3**optical touch sensor**

sensor that senses the change of light intensity by using light emitter and detector pairs arranged around the perimeter of the touch screen

3.5.4**surface acoustic wave touch sensor**

sensor for detecting the change of surface acoustic wave which is generated by the touch input operations of the touch displays

3.6 Input operations of touch displays**3.6.1****touch event**

event of detecting the coordinate of the physical contact with a touch sensor

3.6.2**hovering**

detection which responds to the non-physical touch of a human body or an object which is suspended over the touch sensor

3.6.3**pressure touch**

touch putting pressure on the touch sensor which recognizes the changes in pressure

3.7 Measurement elements of touch displays**3.7.1****touch test bar**

artificial bar that is applied as touch input on the touch screen panel

3.8 Measurement performances of touch displays**3.8.1****false touch**

phenomenon where touch coordinates are incorrectly reported

3.8.2

adjacent touch distance

minimum distance between two separate touch points which the touch screen panel can report as being distinct from one another

3.8.3

maximum hover distance

greatest distance between the hovering input object and the touch screen panel surface for which the object position can be reported

3.8.4

minimum sensing size

smallest diameter of the test bar for which touch data can be reported

3.8.5

touch accuracy

degree which shows how close a measured value is to the targeted (true) value

Note 1 to entry: Unit: mm.

3.8.6

touch jitter

deviation of the reported coordinates from the touch screen panel when a stationary touch is applied

3.8.7

touch latency

delay time from actual touch to reported touch in idle/doze mode

3.8.8

touch linearity

linearity of sensed touch points when a linear dragging touch is applied

3.8.9

touch precision

touch repeatability

degree which shows how close the measured values are to each other with subsequent touches in the same location

3.8.10

touch reporting time

interval time of the touch signal from the touch sensor to the touch controller

3.8.11

move latency

delay time between the actual touch position change and the recognized position change

3.8.12

touch report rate

frequency of touch reports

3.8.13

touch resolution

density (or number per unit length) of distinctly reportable points

3.9 Interactive displays

3.9.1

interactive display

electronic display device that allows the users to interact with the device by touching or applying other non-physical inputs and gives interactive feedback to the users

3.10 Components of interactive displays

3.10.1

interactive display module

combination of interactive screen panel, interactive input sensor, interactive feedback actuator, interactive touch controller, surface cover, and connector that creates a fully integrated module capable of sensing and processing inputs

3.10.2

interactive input sensor

sensor that is connected to the interactive touch controller and recognizes the touch event and interactive inputs

3.10.3

wirelessly input sensor

sensor that wirelessly communicates with the interactive touch controller and recognizes the touch and other interactive inputs

3.10.4

surface input sensor

sensor that is integrated with the display and recognises touch and hovering inputs

3.10.5

optical input sensor

sensor that is connected to the display and recognises proximity and gesture inputs

3.10.6

tactile sensor

sensor that detects the parameters of physical touch of the surface which is touched on the display

3.10.7

hovering sensor

sensor that detects the hovering of an object above the display surface

3.10.8

proximity sensor

sensor that detects the proximity of an object

3.10.9

gesture sensor

sensor that detects the human gesture which is recognized by an optical input sensor and tracking camera sensor

3.10.10

tracking camera sensor

sensor that tracks the movement of objects in space where the camera can recognize the object

3.10.11

interactive feedback actuator

feedback actuator that is integrated with the display and responds to touch and interactive input

3.11 Sensor types of interactive displays

3.11.1

bio-signal input sensor

sensor that detects the signal of human body as input

EXAMPLE Body heat.

3.12 Input operations of interactive displays

3.12.1

finger gesture

interactive input of human gesture that is taken by a combination of finger actions such as finger bending, finger swiping, finger twisting, and finger crossing

3.12.2

palm gesture

interactive input of human gesture that is taken by a combination of palm actions such as palm swiping, palm twisting, and palm fist

3.12.3

arm gesture

interactive input of human gesture that is taken by a combination of arm actions such as arm bending, arm swiping, and arm twisting

3.12.4

gesture recognition

input recognition by interpreting human gestures including finger gesture, palm gesture, and arm gesture

3.12.5

three-dimensional model gesture recognition

gesture recognition that is interpreted by a three-dimensional volumetric modelling of human gesture

3.12.6

skeletal gesture recognition

gesture recognition that is interpreted by a skeletal modelling of human gesture

3.12.7

appearance gesture recognition

gesture recognition that is interpreted by a binary silhouette and contour modelling of human gesture

3.12.8

motion recognition

input recognition by interpreting the movement of an object that is recognized with a tracking camera sensor

3.12.9

eye tracking

interactive input that is recognized with a tracking camera sensor by measuring either the point of gaze or the motion of an eye

3.12.10**head tracking**

interactive input that is recognized with a tracking camera sensor by measuring the motion of a human head

3.13 Actuator types of interactive displays**3.13.1****haptic feedback actuator**

physical actuator that applies force, vibration, and motion to the user as interactive feedback

3.13.2**tactile feedback actuator**

surface tactile actuator that changes surface texture as interactive feedback

3.13.3**electrical feedback actuator**

electrical actuator that applies electrical stimulation to the user as interactive feedback

3.14 Feedback operations of interactive displays**3.14.1****display feedback**

visual feedback which shows interactive images and videos on the interactive screen panel as response to the interactive input

3.14.2**haptic feedback**

physical feedback which applies force, vibration, and motion to the user as response to the interactive input

3.14.3**surface tactile feedback**

tactile feedback which changes the surface texture of the surface cover as response to the interactive input

3.14.4**electrical feedback**

electrical feedback which applies electrical stimulus to the user as response to the interactive input

3.15 Measurement performances of interactive displays**3.15.1****feedback time**

response time between input recognition of the interactive input sensor and interactive feedback

3.15.2**hovering range**

distance range where the surface input sensor recognizes an object as hovering

3.15.3**proximity range**

range of the distance where an optical sensor recognizes an object in proximity

3.15.4

gesture recognition accuracy

accuracy between the actual gesture and the recognized input gesture

3.15.5

tracking speed

maximum speed with which the tracking camera can recognize an object

Annex A (normative)

Structure of touch display module

Figure A.1 shows the structural relationship between the touch sensor, touch controller, touch sensor module, display panel, touch display panel and touch display module.

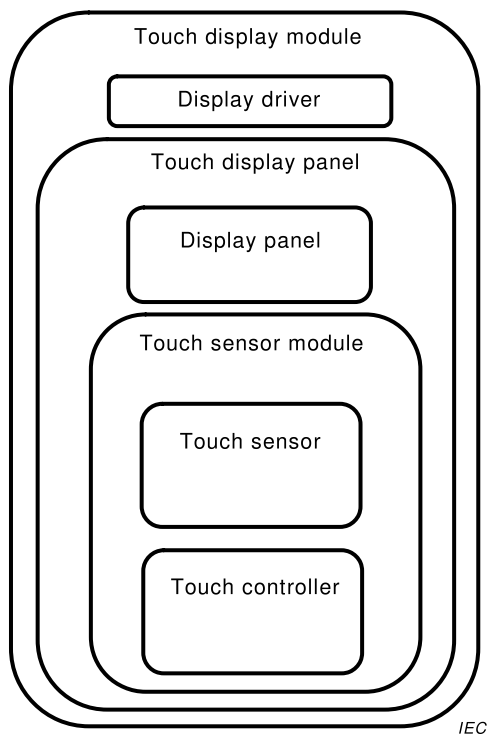


Figure A.1 – Touch display module structure

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